

COLLEGE OF HEALTH SCIENCES

DEPARTMENT OF ANATOMY

***GENERAL HISTOLOGY AND HISTOTECHNIQUE* Exam [Year I]**

Date: November 30, 2013

Time allowed: 3:00 hours.

Name: _____ SCORE _____

For each of the following Multiple Choice Questions, choose the one best appropriate answer and write your answer in BLOCK LETTERS in the spaces provided.

_____ 1. Which of the following **intermediate filaments** is a marker for **connective tissue tumors**?

- a. Keratin b. Desmin c. Vimentin d. Neurofilaments e. Laminins

_____ 2. Which of the following cells is **multinucleated**?

- a. hepatocyte b. neutrophil c. parietal cell d. megakaryocyte e. None of these

_____ 3. Which of the following statements is **false** with regard to **cell morphology**?

- a. Glycolipids are found in the outer leaflet of a plasmalemma only.
b. Lipid bilayer of plasmalemma is freely permeable to charged ions.
c. Lysosomes are particularly abundant in cells exhibiting phagocytic activities.
d. Cholesterol assists in maintaining structural integrity of the cell membrane.
e. Clathrin-coated vesicles are associated with the regulated signal-directed transport of proteins.

_____ 4. Concerning the **cellular morphology**, which statement is **false**?

- a) Centrioles may be located at the base of cilia and flagella. b) Centrioles are membranous organelles.
c) Ribosomes are assembled in the nucleoli. d) Ribosomes are the sites of protein synthesis.
e) The folds on the inner mitochondrial membrane are called cristae.

_____ 5. With regard to the **cellular organelles**, which statement is **true**?

- a) Lysosomal enzymes have optimal activity at basic pH to synthesize rRNA
b) Golgi apparatus possess heterochromatin which is transcriptionally active.
c) Peroxisomes are formed by fusion of phagocytic vacuoles with late endosomes.
d) The nucleolus is a well-defined, membrane-bounded acidophilic nuclear inclusion. e) None of the above.

_____ 6. Concerning structures inside a **cell**, which statement is **false**?

- a. Glycocalyx facilitates cell-cell recognition and interaction.
b. Glycogen synthesis, storage and breakdown in the liver take place in the mitochondria.
c. The nucleolus disperses during the prophase of cell division but reappears in the telophase stage of mitosis.
d. Microvilli are much shorter than stereocilia. e. Microfilaments are more stable than microtubules.

_____ 7. Which one is **not** the function of the **smooth endoplasmic reticulum**?

- a. Membrane retrieval, recycling and redistribution.
b. Steroid hormone synthesis. c. Drug detoxification in hepatocytes.
d. Muscle contraction and relaxation. e. Metabolism of lipid and cholesterol.

_____ 8. **Peroxisomes**

- a. Function in cellular attachment to extracellular matrix components.
b. contain oxidative enzymes. c. are not formed in the Golgi complex.

- d. are the major sites of oxygen utilization. e. are self replicating, like the mitochondria.
9. Which one is **not** the function of the **peroxisomes**?
 a. Regulation of H₂O₂ within cells. b. Gluconeogenesis. c. β -oxidation of long chain fatty acids.
 d. Formation of bile salts. e. None of them f. All of them
10. Which of the following pairs of functions is most closely associated with the **Golgi complex**?
 a) energy metabolism and glycogen synthesis b) energy metabolism and lipid metabolism
 c) glycosylation and sulfation of secretory products d) phagocytosis and receptor recycling
 e) glycogen synthesis and packaging of secretions.
11. How thick is the **plasma membrane**?
 a) 8-10 Ångstroms b) 8-10 nanometers c) 8-10 micrometers d) 8-10 millimeters e). None of the above
12. **Synthesis of all proteins** appears to be initiated on which of the following cellular components?
 a) free polyribosomes b) Golgi complex c) nucleosomes d) RER e) unit membranes
13. **Stereocilia**
 a) are structurally similar to true cilia b) contain axonemes c) are underlain by basal bodies
 d) contain actin filaments in their core e) contain nine microtubule triplets at their core.
14. The location of **Krebs cycle enzyme and mitochondrial DNA** is
 a) cristae b) inner mitochondrial membrane c) matrix granules
 d) mitochondrial matrix e) outer mitochondrial membrane.
15. Which of the following contain abundant **basal mitochondria between infoldings of the basal plasma membrane**?
 a) ion-transporting cells b) mucus-secreting cells c) peptide-secreting cells
 d) protein- secreting cells. e) steroid-secreting cells.
16. Which of the following is **the longest and most variable cell-cycle phase** in continuously dividing cells?
 a) G₀ phase b) G₁ phase c) G₂ phase d) M phase e) S phase
17. **rRNA** is synthesized by which of the following cellular site?
 a) free polyribosomes b) nuclear envelope c) nucleolus d) phagosomes e) RER
18. The **Barr body**,
 a) its DNA replicates during every other cell cycle in normal males.
 b) in normal females, it is a member of homologous chromosome pair.
 c) is composed of euchromatin during the mitotic phase of a cell cycle.
 d) attaches to the inner surface of the plasma membrane during interphase.
 e) occurs when homologous chromosomes fail to separate during an anaphase.
19. Which of the following is **NOT a membranous organelle**?
 a. Lysosomes b. Peroxisomes c. Mitochondria d. Ribosomes e. Endoplasmic reticulum
20. In **pseudostratified columnar epithelium**,
 a) all nuclei lie at the same depth from the surface. b) all cells border onto the lumen
 c) all cells touch the basal lamina. d) cells are always of ectodermal origin. e) all cells are ciliated
21. What forms the **brush border**?
 a. Microvilli b. Stereocilia c. Cilia d. Keratinization e. Mesothelium
22. **Holocrine secretion**,
 a) occurs in endocrine glands. b) involves little or no loss of cytoplasm. c) occurs in sebaceous glands.

d) involves secretion of chlorine or bromine ions. e) is the typical mode of secretion of DNES cells.

____ 23. What type of epithelium is associated with goblet cells?

a. Simple squamous epithelium
d. Stratified squamous epithelium

b. Simple cuboidal epithelium

c. Simple columnar epithelium

e. Pseudostratified epithelium

____ 24. With regard to the **epithelial tissue**, which statement is **false**?

- a. Stereocilia are seen in the epididymis and the hair cells of the ear.
- b. There is very little intercellular space between the cells in epithelial tissue.
- c. Epithelioid tissue is made up of tightly packed epithelial-like cells, but they do not have a free surfaces.
- d. A goblet cell is a unicellular gland.
- e. Compound Tubuloalveolar is another term for an acinar gland.

____ 25. With regard to the **epithelial tissue**, which statement is **true**?

- a) Arteriovenous anastomosis provides nutrition to the highly cohesive rows of cells.
- b) Desmosomes permit the passage of ions between adjacent cells.
- c) Microvilli contain sheaf of microtubules with a 9+2 axoneme pattern.
- d) The major constituent of epithelia is abundant extracellular matrix.
- e) Merocrine glands are the most common types of exocrine glands.

____ 26. With regard to the **epithelial tissue**, which statement is **false**?

- a) The epidermis is formed by stratified squamous keratinized epithelium.
- b) Mesothelium is simple squamous epithelium that lines the vascular system.
- c) The lining of the gastrointestinal tract is simple columnar epithelium.
- d) An example of epithelioid tissue is the parenchyma of the adrenal gland.
- e) None of the above.

____ 27. One of the following structures is lined by **stratified columnar epithelium**.

- a) Urinary bladder.
- b) Conjunctiva.
- c) Esophagus.
- d) Vagina.
- e) Trachea.

____ 28. One of the following structures is **NOT** lined by **stratified squamous nonkeratinized epithelium**.

- a) esophagus
- b) vagina
- c) anus
- d) oral cavity
- e) ureter

____ 29. Which of the following describes a **secretory process** in which no cell membrane components or **cytosolic contents are lost**?

- a) Apocrine
- b) Merocrine
- c) Holocrine
- d) Endocrine
- e) All of them

____ 30. **Zonula occludens**,

- a) surrounds the entire columnar cells in the basal region of their lateral plasma membranes.
- b) is characterized by the presence of abundant cytokeratin filaments in the vicinity of the junction.
- c) is characterized by the presence of connexons that link the plasma membranes of adjacent cells.
- d) is characterized by a dense intracellular plaque.
- e) None of the above.

____ 31. Which of the following **cell types** produces and maintains all **extracellular matrix** components of connective tissue?

- a) plasma cell
- b) mast cell
- c) mesenchymal cell
- d) macrophage
- e) fibroblast

____ 32. Which of the following is **NOT** primarily composed of connective tissue?

- a. Blood
- b. Bone
- c. Tendon
- d. Intervertebral disc
- e. Myometrium

____ 33. Which of the following is **NOT** a fiber found in connective tissue?

- a. Collagen fiber
- b. Elastic fiber
- c. Reticular fiber
- d. Purkinje fiber
- e. All of the above are fibers found in connective tissue

____ 34. Which **connective tissue** cell type contains **properties of smooth muscle cells**?

- a. Fibroblast
- b. Myofibroblast
- c. Histiocyte
- d. Plasma cell
- e. Mast cell

- _____ 35. Which of the following cells characteristically **synthesize and secrete immunoglobulins**?
a) fibroblasts b) lymphocytes c) macrophages d) mast cells e) plasma cells
- _____ 36. Which of the following contain **secretory granules** filled with **heparin and histamine**?
a) fibroblasts b) lymphocytes c) macrophages d) mast cells e) plasma cells
- _____ 37. Which of the following are the three **basic components of all types of connective tissue**?
a) arteries, veins, and capillaries b) cells, fibers, and ground substances
c) mast cells, lymphocytes and adipocytes. d) type-II collagen, hyaluronic acid, and fibronectin.
e) fibroblasts, reticular fibers, and proteoglycan aggregates.
- _____ 38. Which of the following would be best suited to visualize reticular fibers?
a. Wright's stain b. Hematoxylin and eosin stain c. Sudan stain
d. Silver impregnation e. Masson's trichrome stain
- _____ 39. With regard to **fibroblasts**, which statement is **false**?
a. produce collagen. b. are tissue macrophages. c. are connective tissue cells derived from mesenchyme
d. are routinely seen in loose connective tissue e. produce the ground substance in connective tissue.
- _____ 40. Which of the following contain the **most lysosomes**?
a) fibroblasts b) lymphocytes c) macrophages d) mast cells e) plasma cells
- _____ 41. Which of the following **macrophages as mononuclear phagocytes** is **false**?
a. Kupffer cells are the hepatic macrophages. b. Langerhans cells are macrophages seen in the skin.
c. Dust cells are alveolar macrophage found in the respiratory tract.
d. Microglia are the central nervous system macrophages. e. None of the above f. All of the above
- _____ 42. Beauty treatment for the reduction of wrinkles is the injection of **hyaluronic acid** into the wrinkle.
What is hyaluronic acid?
a. Dermatan sulfate b. Proteoglycan c. Glycosaminoglycan d. Chondroitin sulfate e. Keratan sulfate
- _____ 43. Which of the following statements about **connective tissue** is **false**?
a) Tendons and ligaments are composed of dense regular connective tissue.
b) areolar tissue is loose irregular connective tissue.
c) Collagen fibers are the most abundant fiber type in connective tissue.
d) Chondroitin sulfate, keratan sulfate, and hyaluronic acid are all glycosaminoglycans. e) None of the above.
- _____ 44. Which of the following can be classified as "**connective tissue proper**"?
a. Adipose tissue b. Dense irregular connective tissue c. Bone d. Blood e. Cartilage
- _____ 45. What type of tissue makes up the **dermis of the skin**?
a. Mucous connective tissue b. Mesenchyme c. Loose irregular connective tissue
d. Dense irregular connective tissue e. Dense regular connective tissue
- _____ 46. Which of the following is **NOT** associated with **connective tissue**?
a. Tightly packed cells b. Extracellular fibers c. Tissue fluid
d. Ground substance e. None of the above; all of the above are seen with connective tissue
- _____ 47. What type of **adipose tissue** tends to increase as humans age?
a. Brown adipose tissue b. White adipose tissue c. Unilocular adipose tissue
d. Multilocular adipose tissue e. Both b and c
- _____ 48. Select the **correct statement about connective tissue**.
a. Dense connective tissue is more abundant than loose connective tissue.

- b. Elastic cartilage is the most abundant type of cartilage.
c. Fibrocartilage is present in the temporomandibular joint.
d. Proteoglycan is glycosaminoglycan in a cartilage. e. Epiglottis is composed of hyaline cartilage.

49. Concerning the **cartilage**, which statement is **true**?

- a) Fibrocartilage is found in the epiphyses of the long bones in young children.
b) Cartilage harbors hematopoietic organs. c) The collagen in elastic cartilage matrix is mainly type IV
d) Elastic cartilage is the primary skeletal tissue in the fetus. e) articular cartilage lacks a perichondrium.

50. Which of the following occurs first in **cartilage histogenesis**?

- a) appositional growth b) differentiation of chondroblasts into chondrocytes
c) formation of mesenchymal condensations d) interstitial growth e) secretion of the matrix

51. What is the **mature cell in cartilage** called?

- a. Chondrocyte b. Chondroblast c. Osteocyte d. Osteoclast e. Bone lining cell

52. Which type of **cartilage** is characterized by the presence of **chondrocytes sitting in lacunae**?

- a. Hyaline cartilage b. Elastic cartilage c. Fibrocartilage d. All of them e. None of them

53. Which **type of cartilage** forms the hammer, anvil and stirrup?

- a. Hyaline cartilage b. Elastic cartilage c. Fibrocartilage d. All of the above e. None of the above

54. Which type of **cartilage** is highly vascular?

- a. Hyaline cartilage b. Elastic cartilage c. Fibrocartilage d. All of the above e. None of the above

55. **Hyaline cartilage** is not found in the

- a) skeleton of the fetus. b) epiphyseal growth plate. c) articular surface on bones.
d) rings of the trachea e) epiglottis

56. There is **no fibrocartilage** in the

- a) larynx b) symphysis pubis. c) intervertebral disc.
d) temporomandibular joint e) All of them a) None of them

57. Concerning the **cartilage**, which statement is **true**?

- a) Fibrocartilage is found in the epiphyses of the long bones in young children.
b) Cartilage harbors hematopoietic organs. c) The collagen in elastic cartilage matrix is mainly type IV.
d) Elastic cartilage is the primary skeletal tissue in the fetus. e) articular cartilage lacks a perichondrium.

58. Which of the following **cell types in bone** is derived from **blood monocytes**?

- a) endothelial cell b) fibroblast c) osteoblast d) osteoclast e) osteocyte.

59. **Primary bone**,

- a) is also termed lamellar bone. b) may appear as either spongy or compact bone.
c) is characterized by the presence of Volkmann's canal. d) has a higher mineral content than the secondary bone.
e) is the first bone tissue to appear during both intramembranous and endochondral bone formation.

60. Which of the following **impedes the distribution of nutrients and oxygen to osteocytes**?

- a) bone matrix b) canaliculi c) filopodia d) Haversian canals e) periosteum.

61. Which of the following hormones is most responsible for increasing **osteoclast** number and activity?

- a) ADH b) calcitonin c) estrogen d) parathyroid hormone e) testosterone.

62. Select the **correct statement**.

- a) Osteocyte is involved in laying down new bone. b) Howship's lacuna is unmineralized bone matrix.

- c) Endosteum is attached to the surface of bone by Sharpey's fibers.
- d) Compact (dense) bone is characterized by the presence of osteons.
- e) Dense regular connective tissue is abundantly found in the bone matrix.

_____ 63. With regard to **bone morphology**, which statement is **false**?

- a. Spicules are the unmineralized tissue in spongy bone.
- b. An osteoclast is a bone resorbing cell.
- c. The osteoblast is the bone forming cell.
- d. Osteoid is unmineralized bone matrix.
- e. Trabeculae are the spicules seen with spongy bone.

_____ 64. Which type of bone has **spicules**?

- a. Immature bone
- b. Dense bone
- c. Compact bone
- d. Cancellous bone
- e. Woven bone

_____ 65. What is the primary component of **red marrow**?

- a. Hematopoietic tissue
- b. Fat
- c. Cartilage
- d. Fibrous tissue
- e. Bone

9. Which cell is an **immature bone cell**?

- a. Osteoclast
- b. Osteon
- c. Osteocyte
- d. Osteoblast
- e. Osteoid

_____ 66. What is in the **bone matrix**?

- a. Elastic fibers
- b. Collagen fibers
- c. Reticular fibers
- d. Dense irregular connective tissue
- e. Dense regular connective tissue

_____ 67. What is **immature bone**?

- a. Dense bone
- b. Woven bone
- c. Cancellous bone
- d. Compact bone
- e. Spongy bone

_____ 68. Concerning the **bone**, which statement is **false**?

- a) Osteoblasts synthesize and secrete osteon.
- b) Osteoprogenitor cells are capable of differentiating into osteoblasts.
- c) Osteoprogenitor cells are located in the periosteum and endosteum.
- d) Osteoprogenitor cells may change into chondrogenic cells at low oxygen tensions.
- e) Chondroclasts found in eroded regions of epiphyseal calcified cartilage are similar in shape to osteoclasts.

_____ 69. With regard to **bone**, which statement is **true**?

- a) Periosteum is a layer of dense connective tissue surrounding the bone.
- b) As the osteoclasts become trapped in their own calcified deposits, they are known as osteoblasts.
- c) Primary bone is remodelled and replaced by secondary bone in tooth sockets and near suture lines in skull bones.
- d) The ruffled borders of osteocytes are irregular finger-like cytoplasmic projections extending into Howship's lacunae.
- e) osteoclast activity is hormonally regulated, stimulated by calcitonin from C-cells of the thyroid and inhibited by parathyroid hormone.

_____ 1. When using a **compound microscope**, objective lenses can be found to have a magnification of all of the following, **EXCEPT**?

- a. 4X
- b. 10X
- c. 40X
- d. 100X
- e. 1000X

_____ 2. What is the total **magnification** achieved with a compound microscope?

- a. Magnification of objective lens
- b. Magnification of ocular lens
- c. Magnification of ocular lens added to the magnification of the objective lens
- d. Magnification of ocular lens multiplied by the magnification of the objective lens
- e. Magnification of condenser lens multiplied by the magnification of the objective lens

_____ 3. Select the **incorrect** statement.

- a. Resolving power is the ability to see two things as discrete images.
- b. The diaphragm controls the diameter of the cone of light.
- c. Parafoal is the term used for a microscope if the focus knob does not have to be readjusted when changing the magnifications.

d. Phase contrast microscope would be particularly useful for looking at living cells.

e. None of the above.

____ 4. Select the **incorrect** statement.

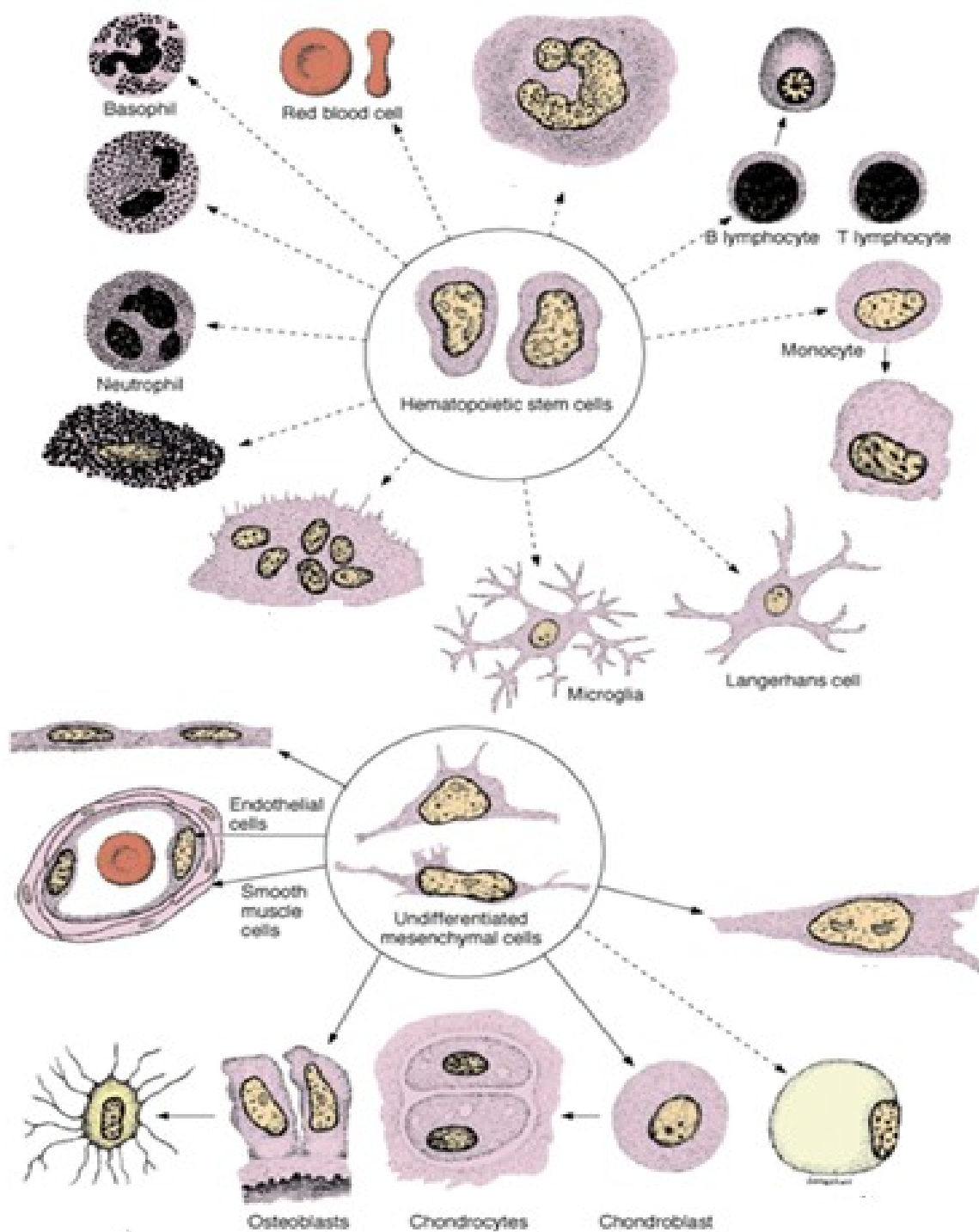
- a. Compound microscope is another name for the light microscope.
- b. The eyepiece lens improve resolution.
- c. The objective lenses enlarge and project the illuminated image of the object in the direction of the eyepiece.
- d. The maximal resolving power of the light microscope is approximately 0.2 μm ;
- e. The resolving power of a microscope depends mainly on the quality of its objective lens.

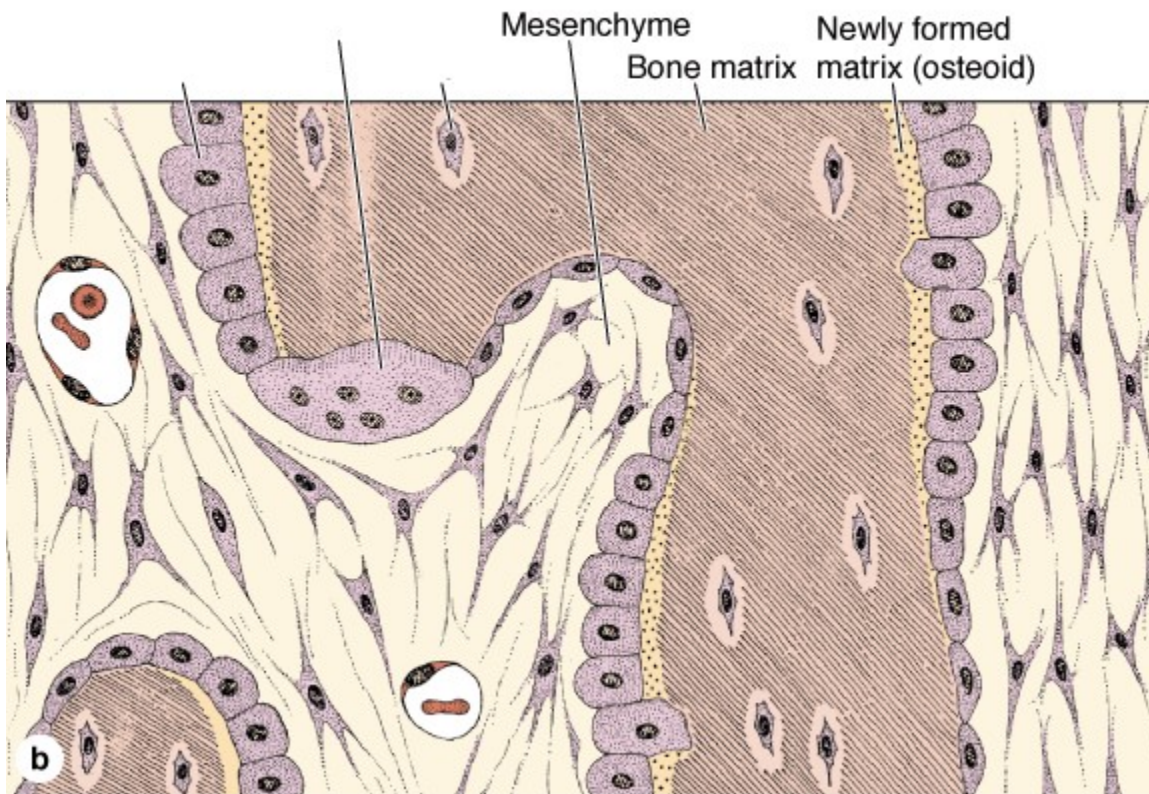
____ 5. What is the role of the **condenser lens**?

- a. Control the aperture of light
- b. Increase the magnification
- c. Focus the light on the specimen
- d. Initial magnification of 10X
- e. Provide light

Label the diagrams

I.





Distribution and main functions of the cells of the mononuclear phagocyte system.

PART II. Answer the following questions.

1. Describe the cell cycle.
2. Enumerate the functions of
 - a. Epithelial tissue.
 - b. Connective tissue.
3. What roles do vitamins and hormones play in the growth of bones?
Emphasis: excess and deficiency.

PART III.

HISTOTECHNIQUE

[30 points]

1. Formulae and their description

[2 points]

a. Numerical aperture

b. Resolution

2. In any one of the objectives of a light microscope, the following numbers may be engraved. What do they represent?

[2 points]

40 _____

0.75 _____

160 _____

0.17 _____

3. Enumerate the optical components of a light microscope and their functions.

[3 points]

4. Distinguish the differences between real image and virtual image.

[2 points]

5. What are the basic histochemical principles?

[4 points]

6. How would you demonstrate the presence of the following in tissue sections?

[9 points]

a. Iron

b. DNA

c. Phosphatates

7. How are antibodies labeled?

[2 points]

8. How would you isolate proteins?

[2 points]

9. Distinguish the differences between direct and indirect methods of immunocytochemistry. [4 points]

